Computational Finance Assignment 1

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1 Derivatives Pricing

a. Discretely Monitored Asian Option

Consider a discretely monitored Asian option with the payoff at maturity T defined as:

$$\Lambda_T = \max\left(\prod_{n=1}^{N} S_{t_n}^{1/N} - K, 0\right),$$

where $\{t_n\}_{n=1}^{N}$ are equally spaced with $t_n = n \cdot \Delta t$ and $N \cdot \Delta t = T$.
Note that:

$$\prod_{i=1}^{n} S_{t_i} = S_{t_n} \cdot \left(\frac{S_{t_{n-1}}}{S_{t_{n-2}}}\right)^2 \cdot \left(\frac{S_{t_{n-2}}}{S_{t_{n-3}}}\right)^3 \cdots \left(\frac{S_{t_1}}{S_{t_0}}\right)^n \cdot S_0^n.$$

We further assume a log-normal model for the asset price:

$$S_{t_{n+1}} = S_{t_n} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_n\right),$$

for i.i.d. $Z_n \sim \mathcal{N}(0,1)$.
Find the distribution of $\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right)$.

We start by iterating from $i = 0$ to $i = n$ the log-normal model for the asset price S_{t_n}:

$$\begin{aligned}
S_{t_n} &= S_{t_{n-1}} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_{n-1}\right) \\
&= S_{t_{n-2}} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_{n-2}\right) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_{n-1}\right) \\
&= S_{t_{n-2}} \prod_{j=n-2}^{n-1} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right) \\
&= S_{t_{n-3}} \prod_{j=n-3}^{n-1} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right) = \cdots =
\end{aligned}$$

$$= S_0 \prod_{j=0}^{n-1} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right).$$

Now, we continue by computing $\ln(S_{t_n}/S_0)$:

$$\begin{aligned}
\ln(S_{t_n}/S_0) &= \ln\left(S_0 \prod_{j=0}^{n-1} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right)/S_0\right) \\
&= \ln\left(\prod_{j=0}^{n-1} \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right)\right) \\
&= \sum_{j=0}^{n-1} \ln\left(\exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right)\right) \\
&= \sum_{j=0}^{n-1} \left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right) \\
&= \left(\mu - \frac{1}{2}\sigma^2\right) n\Delta t + \sigma\sqrt{\Delta t} \sum_{j=0}^{n-1} Z_j.
\end{aligned}$$

Then, we can simplify $\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right)$ as

$$\begin{aligned}
\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right) &= \ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}\right) - \ln(S_0) = \sum_{n=1}^{N} \ln(S_{t_n}^{1/N}) - \ln(S_0) \\
&= \frac{1}{N}\sum_{n=1}^{N} \ln(S_{t_n}) - \ln(S_0) = \frac{1}{N}\sum_{n=1}^{N} \ln(S_{t_n}/S_0).
\end{aligned}$$

Substituting the expression we found for $\ln(S_{t_n}/S_0)$ into $\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right)$, we get

$$\begin{aligned}
\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right) &= \frac{1}{N}\sum_{n=1}^{N}\left(\left(\mu - \frac{1}{2}\sigma^2\right) n\Delta t + \sigma\sqrt{\Delta t}\sum_{j=0}^{n-1} Z_j\right) \\
&= \left(\mu - \frac{1}{2}\sigma^2\right)\frac{(N+1)\Delta t}{2} + \frac{\sigma\sqrt{\Delta t}}{N}\sum_{n=1}^{N}\sum_{j=0}^{n-1} Z_j,
\end{aligned}$$

where we have used that $\sum_{i=1}^{n} i = n(n+1)/2$. Now, we need to compute the value of $\sum_{n=1}^{N}\sum_{j=0}^{n-1} Z_j$. Note that for a fixed n, we are summing all Z_j such that $0 \le j \le n-1$ and $1 \le n \le N$. To simplify this, instead of fixing n we will fix j. Observe that fixing j, implies summing the values Z_j for which $j \le n-1$. This inequality becomes $n \ge j+1$ and, as $n \le N$, then, these are the values $n \in \{j+1, j+2, \ldots, N\}$. In total we have $(N-j)$ of that n-values and, as a

consequence, each Z_j appears exactly $(N-j)$ times. Given this, we can rewrite the previous expression as follows:

$$\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right) = \left(\mu - \frac{1}{2}\sigma^2\right)\frac{(N+1)\Delta t}{2} + \frac{\sigma\sqrt{\Delta t}}{N}\sum_{j=0}^{N-1}(N-j)Z_j. \tag{1}$$

Now, we use the fact that $Z_n \sim N(0,1)$, that they are i.i.d, and that a linear combination of normal distributions is a normal distribution. Then, we need to find its mean and variance. The mean is

$$\begin{aligned}
\mathbb{E}\left[\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right)\right] &= \mathbb{E}\left[\left(\mu - \frac{1}{2}\sigma^2\right)\frac{(N+1)\Delta t}{2} + \frac{\sigma\sqrt{\Delta t}}{N}\sum_{j=0}^{N-1}(N-j)Z_j\right] \\
&= \mathbb{E}\left[\left(\mu - \frac{1}{2}\sigma^2\right)\frac{(N+1)\Delta t}{2}\right] \\
&= \left(\mu - \frac{1}{2}\sigma^2\right)\frac{(N+1)T}{2N},
\end{aligned}$$

where in the second equality we have used that the mean of Z_i is 0 $\forall i$, and in the third equality that $\Delta t = T/N$. Next, we compute the variance:

$$\begin{aligned}
\mathbb{V}\left[\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right)\right] &= \mathbb{V}\left[\left(\mu - \frac{1}{2}\sigma^2\right)\frac{(N+1)\Delta t}{2} + \frac{\sigma\sqrt{\Delta t}}{N}\sum_{j=0}^{N-1}(N-j)Z_j\right] \\
&= \mathbb{V}\left[\frac{\sigma\sqrt{\Delta t}}{N}\sum_{j=0}^{N-1}(N-j)Z_j\right] = \frac{\sigma^2 T}{N^3}\mathbb{V}\left[\sum_{j=0}^{N-1}(N-j)Z_j\right] \\
&= \frac{\sigma^2 T}{N^3}\sum_{j=0}^{N-1}(N-j)^2 = \frac{\sigma^2 T}{N^3}\sum_{k=1}^{N}k^2 \\
&= \frac{\sigma^2 T}{N^3}\frac{N(N+1)(2N+1)}{6} \\
&= \frac{\sigma^2 T(N+1)(2N+1)}{6N^2},
\end{aligned}$$

where in the fourth equality we have used that the variance of Z_i is 1 $\forall i$. In conclusion, we have deduced that the distribution of $\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right)$ is the following:

$$\ln\left(\prod_{n=1}^{N} S_{t_n}^{1/N}/S_0\right) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right)\frac{(N+1)T}{2N}, \frac{\sigma^2 T(N+1)(2N+1)}{6N^2}\right).$$

b. Continuously Monitored Asian Option PDE

Now consider a continuously monitored Asian option with payoff at maturity T defined as $\Lambda_T = \max(A_T - K, 0)$, where A_T is the time-T value of the process:

$$A_t = \frac{1}{T}\int_0^t S_u\, du, \quad t \in [0, T].$$

We price the option under the Black-Scholes model in the risk-neutral measure:

$$dS_t = rS_t\, dt + \sigma S_t\, dB_t,$$

where:

- S_t is the stock price,
- r is the constant risk-free rate,
- σ is the constant volatility,
- B_t is a standard Brownian motion.

Let C_t denote the price of the continuously monitored Asian call at time t. Derive the PDE that C_t satisfies. Specify all boundary conditions.

First, we compute the differential of A_t, which gives us $dA_t = \frac{1}{T}S_t dt$. Then, one can observe that the price of the continuously monitored Asian call depends on time, on the stock price S_t and on the time-T value of the process A_t and, consequently, we can write $C = C(t, S_t, A_t)$. Now, we compute the differential of C using Ito's Lemma:

$$\begin{aligned}
dC(t, S_t, A_t) &= \frac{\partial C}{\partial t}dt + \frac{\partial C}{\partial S_t}dS_t + \frac{\partial C}{\partial A_t}dA_t + \frac{1}{2}\frac{\partial^2 C}{\partial S_t^2}(dS_t)^2 + \frac{1}{2}\frac{\partial^2 C}{\partial A_t^2}(dA_t)^2 + \frac{1}{2}\frac{\partial^2 C}{\partial t^2}(dt)^2 \\
&+ \frac{\partial^2 C}{\partial S_t \partial A_t}(dS_t)(dA_t) + \frac{\partial^2 C}{\partial t \partial S_t}(dt)(dS_t) + \frac{\partial^2 C}{\partial t \partial A_t}(dt)(dA_t) \\
&= \frac{\partial C}{\partial t}dt + \frac{\partial C}{\partial S_t}dS_t + \frac{\partial C}{\partial A_t}dA_t + \frac{1}{2}\frac{\partial^2 C}{\partial S_t^2}(dS_t)^2 \\
&= \frac{\partial C}{\partial t}dt + \frac{\partial C}{\partial S_t}\left(rS_t\, dt + \sigma S_t\, dB_t\right) + \frac{\partial C}{\partial A_t}(\frac{1}{T}S_t dt) \\
&+ \frac{1}{2}\frac{\partial^2 C}{\partial S_t^2}\sigma^2 S_t^2 dt \\
&= \left(\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S_t}rS_t\frac{1}{T}\frac{\partial C}{\partial A_t}S_t + \frac{1}{2}\frac{\partial^2 C}{\partial S_t^2}\sigma^2 S_t^2\right)dt + \frac{\partial C}{\partial S_t}\sigma S_t dB_t.
\end{aligned}$$

Using the no-arbitrage principle, we know that in the Black-Scholes model in the risk-neutral model, the discounted option price is a martingale under the risk-free measure $\mathbb{Q}$ (i.e. $e^{-rt}C(t, S_t, A_t)$ must be a martingale). Then, we

compute the differential of $e^{-rt}C(t, S_t, A_t)$ to find its drift term, which must be 0, so $e^{-rt}C(t, S_t, A_t)$ is a martingale.

$$\begin{aligned}
d(e^{-rt}C(t, S_t, A_t)) &= -re^{-rt}Cdt + e^{-rt}dC \\
&= e^{-rt}\left(\left(\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S_t} rS_t \frac{1}{T}\frac{\partial C}{\partial A_t} S_t + \frac{1}{2}\frac{\partial^2 C}{\partial S_t^2}\sigma^2 S_t^2\right) dt + \frac{\partial C}{\partial S_t}\sigma S_t dB_t\right) \\
&- re^{-rt}Cdt \\
&= e^{-rt}\left(\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S_t} rS_t \frac{1}{T}\frac{\partial C}{\partial A_t} S_t + \frac{1}{2}\frac{\partial^2 C}{\partial S_t^2}\sigma^2 S_t^2 - rC\right) dt \\
&+ e^{-rt}\frac{\partial C}{\partial S_t}\sigma S_t dB_t.
\end{aligned}$$

Now, for the drift term to be 0, which is the term which multiplies dt, we impose the following:

$$\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S_t} rS_t + \frac{1}{T}\frac{\partial C}{\partial A_t} S_t + \frac{1}{2}\frac{\partial^2 C}{\partial S_t^2}\sigma^2 S_t^2 - rC = 0,$$

that is the PDE that the Asian call price $C(t, S_t, A_t)$ satisfies for $(t, S_t, A_t) \in [0, T) \times (0, \infty) \times [0, \infty)$. Then, we must define the boundary conditions. First, we know that the price of the Asian option at the maturity time T must be $\Lambda_T = \max(A_T - K, 0)$. Therefore, we write $C(T, S_T, A_T) = \max(A_T - K, 0)$. Moreover, when $S \to 0$, the asset price goes to 0, and it stays at 0. The time-T value of the process remains the current average, this is $A_t = A$. Then, the option payoff is known and we discount it at the risk-free rate, which gives us $C(t, 0, A) = e^{-r(T-t)}\max(A - K, 0)$.

c. Feynman-Kac Formulation

To simplify the calculation, assume $S_t = e^{B_t}$ (where B_t is a standard Brownian motion with $B_0 = 0$) and define $A_T = \int_0^T S_t\, dt$. Let:

$$\varphi(w) = \mathbb{E}\left[e^{iwA_T}\right], \quad w \in \mathbb{R}.$$

Use the Feynman-Kac theorem to formulate $\varphi(w)$ as a solution to a PDE, together with the appropriate boundary and terminal conditions.

Observe that we have $S_t = e^{B_t}$, with B_t a standard Brownian motion with $B_0 = 0$, and $A_T = \int_0^T S_t dt = \int_0^T e^{B_t} dt$. Our objective is to compute $\varphi(w) = \mathbb{E}\left[e^{iwA_T}\right]$ with $w \in \mathbb{R}$. To show this, we use the Feynman-Kac formula from the slides, which states that

$$V(x, t) = \mathbb{E}\left[e^{-\int_t^T r(X_u, u)du} g(X_T) \mid X_t = x\right],$$

where X_s solve the diffusion process

$$dX_s = b(X_s, s)ds + \sigma(X_s, s)dW_s, \quad s \geq t,$$

with initial condition $X_t = x$, and where $V(x,t)$ is the solution of the PDE

$$\frac{\partial V}{\partial t} + b(x,t)\frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2(x,t)\frac{\partial^2 V}{\partial x^2} - r(x,t)V = 0,$$

with terminal condition $V(x,T) = g(x)$. Notice that we can write e^{iwA_t} in integral form:

$$e^{iwA_T} = e^{iw\int_0^T e^{B_s}ds} = e^{-\int_0^T(-iwe^{B_s})ds}.$$

Consequently, if we define $v(x,t) = \mathbb{E}\left[e^{-iw\int_t^T e^{B_s}ds} \mid B_t = x\right]$, we get that $\varphi(w) = v(0,0)$, and therefore we just need to determine the values b, σ and r to apply the Feynman-Kac formula. In first place, one observes that $g(X_T) = 1$. Also, one notices that $X_t = B_t$, which implies that $dX_t = dB_t$, hence $b(x,t) = 0$ and $\sigma(x,t) = 1$. Moreover, one can deduce that $r(x,t) = -iwe^x$. Using all these conditions, and that $v(x,t)$ must satisfy the PDE described by the Feynman-Kac theorem, we get that:

$$\frac{\partial v}{\partial t} + \frac{1}{2}\frac{\partial^2 v}{\partial x^2} + iwe^x v = 0,$$

with terminal condition $v(x,T) = g(x) = 1\ \forall x \in \mathbb{R}$, and $\varphi(w) = v(0,0)$.

2 The COS method

Follow the steps in Section 2.1 "Inverse Fourier integral via cosine expansion" of the original COS paper (available on BrightSpace under Lecture 7).

a. Sine Series Expansion
Replace equations (5) and (6) of the COS paper by a **Fourier-sine series expansion** instead, to derive a **sine version** of equation (11) of the paper.

In the original COS paper, a function $f(x)$ supported on $[a, b]$ is approximated using a Fourier cosine series:

$$f(x) = \sum_{k=0}^{\infty}{}' A_k \cos\left(k\pi \frac{x-a}{b-a}\right) \quad \text{where} \quad A_k = \frac{2}{b-a}\int_a^b f(x)\cos\left(k\pi\frac{x-a}{b-a}\right)dx,$$

where $'$ denotes that the first term in the summation is weighted by one-half. For the Fourier sine series, we use that $\sin(0) = 0$, so the first term of the summation will not be weighted by one-half, and the summation will start with $k = 1$. Then, we can write

$$f(x) = \sum_{k=1}^{\infty} B_k \cos\left(k\pi\frac{x-a}{b-a}\right) \quad .$$

Now, we use the orthogonality condition, which states that

$$\int_a^b \sin\left(k\pi\tfrac{x-a}{b-a}\right)\sin\left(j\pi\tfrac{x-a}{b-a}\right)dx = \tfrac{b-a}{2}\delta_{kj}.$$

Then, we define the coefficients B_k as follows:

$$B_k = \frac{2}{b-a}\int_a^b f(x)\sin\left(k\pi\frac{x-a}{b-a}\right)dx.$$

Following the paper's argument, we suppose that $[a, b] \in \mathbb{R}$ is chosen such that the truncated integral approximates the infinite counterpart very well, i.e.,

$$\phi(w) = \int_{\mathbb{R}} e^{iwx} f(x)dx \approx \int_a^b e^{iwx} f(x)dx = \phi_1(w). \tag{2}$$

Recalling Euler's formula, which states that $e^{i\theta} = \cos(\theta) + i\sin(\theta)$, we use it to rewrite the sin term:

$$\exp\left(ik\pi\frac{x-a}{b-a}\right) = \cos\left(k\pi\frac{x-a}{b-a}\right) + i\sin\left(k\pi\frac{x-a}{b-a}\right),$$

therefore, we deduce that

$$\sin\left(k\pi\frac{x-a}{b-a}\right) = \Im\left(\exp\left(ik\pi\frac{x-a}{b-a}\right)\right)$$

$$= \Im\left(\exp\left(ik\pi\frac{x}{b-a}\right)\exp\left(-ik\pi\frac{a}{b-a}\right)\right),$$

where $\Im(\cdot)$ denotes the imaginary part of the argument. Then, we can write B_k as

$$B_k \equiv \frac{2}{b-a}\Im\left(\phi_1\left(\frac{k\pi}{b-a}\right)\exp\left(-i\frac{k\pi a}{b-a}\right)\right),$$

where we have used that $w = \frac{k\pi}{b-a}$. Now, employing (2), we can write $F_k \approx B_k$ as follows:

$$F_k \equiv \frac{2}{b-a}\Im\left(\phi\left(\frac{k\pi}{b-a}\right)\exp\left(-i\frac{k\pi a}{b-a}\right)\right).$$

Replacing B_k by F_k in the series expansion of $f(x)$ on $[a, b]$, we have

$$f_1(x) = \sum_{k=1}^{\infty} F_k \sin\left(k\pi\frac{x-a}{b-a}\right),$$

and truncating the series summation we obtain

$$f_2(x) = \sum_{k=1}^{N} F_k \sin\left(k\pi\frac{x-a}{b-a}\right),$$

which is the sine version of equation (11) in the paper.

b. Numerical Verification

Program your sine version of equation (11) in Python and reproduce the **errors** in Table 1 of the paper (copied below). Note that only the errors are needed, not the CPU times. *(Table 1 from the original COS paper to be included by course staff.)*

We implement the code and show *Table 1*:

N	Max Absolute Error
4	2.107e-01
8	8.258e-02
16	4.634e-03
32	1.791e-07
64	1.042e-16

Figure 1: Maximum error when recovering $f(x)$ from $\phi(w)$ by Fourier-sine expansion

Notice that the errors are similar to the ones from the Fourier-cosine expansion.

c. Sine Version Pricing Formula for European Put Options

Derive the sine version of the pricing formula for European put options, analogous to equations (19) and (25) of the original COS paper.

Hint: The characteristic function of the log-asset price $x = \ln(S_T/K)$ is typically known. Following the same convention as in the paper, express the option payoff as a function of the log-asset price x.

We start with the risk-neutral valuation formula, which is given by

$$v(x, t_0) = e^{-r\Delta t}\mathbb{E}^{\mathbb{Q}}[v(y, T)|x] = e^{-r\Delta t}\int_{\mathbb{R}} v(y, T)f(y|x)dy,$$

where v denotes the option value, Δt is the difference between the maturity, T, and the initial date, t_0, and $\mathbb{E}^{\mathbb{Q}}[\cdot]$ is the expectation operator under risk-neutral measure $\mathbb{Q}$. x and y are state variables at times t_0 and T, respectively; $f(y|x)$ is the probability density of y given x, and r is the risk-neutral interest rate. Then, similar to the COS paper, since the density rapidly decays to zero as $y \to \pm\infty$, we truncate the infinite integration rage without losing significant accuracy to $[a, b] \subset \mathbb{R}$, and we obtain approximation v_1:

$$v_1(x, t_0) = e^{-r\Delta t}\int_a^b v(y, T)f(y|x)dy.$$

In the second step, since $f(y|x)$ is usually not known whereas the characteristic function is, we replace the density by its cosine expansion in y,

$$f(y|x) = \sum_{k=1}^{+\infty} B_k(x)\sin\left(k\pi\frac{y-a}{b-a}\right),$$

with

$$B_k(x) := \frac{2}{b-a}\int_a^b f(y|x)\sin\left(k\pi\frac{y-a}{b-a}\right)dy,$$

so that

$$v_1(x, t_0) = e^{-r\Delta t}\int_a^b v(y, T)\sum_{k=1}^{+\infty} B_k(x)\sin\left(k\pi\frac{y-a}{b-a}\right)dy.$$

Now, we interchange the summation and integration, and we insert the definition

$$V_k := \frac{2}{b-a}\int_a^b v(y, T)\sin\left(k\pi\frac{y-a}{b-a}\right)dy,$$

resulting in

$$v_1(x,t_0) = \frac{1}{2}(b-a)e^{-r\Delta t} \cdot \sum_{k=1}^{+\infty} B_k(x)V_k.$$

Note that the V_k are the sine series coefficients of payoff function $v(y,T)$ in y. In this process, we have transformed the product of two real functions, $f(y|x)$ and $v(y,T)$, into that of their Fourier-sine series coefficients. Due to the rapid decay rate of these coefficients, we further truncate the series summation to obtain approximation v_2:

$$v_2(x,t_0) = \frac{1}{2}(b-a)e^{-r\Delta t} \cdot \sum_{k=1}^{N} B_k(x)V_k.$$

Similar to 2(a), coefficients B_k can be approximated by $F_k(x)$. Replacing $B_k(x)$ by $F_k(x)$ in the expression for v_2, we obtain

$$v(x,t_0) \approx v_3(x,t_0) = e^{-r\Delta t} \sum_{k=1}^{N} \Im\left(\phi\left(\frac{k\pi}{b-a};x\right)\exp\left(-i\frac{k\pi a}{b-a}\right)\right)V_k,$$

with characteristic function ϕ.

Now, as done in the paper, before we can use v_3 for pricing options, the payoff series coefficients, V_k, have to be recovered. As we assume here that the characteristic function of the log-asset price is known, we represent the payoff as a function of the log-asset price. Let us denote the log-asset prices by $x := \ln(S_0/K)$ and $y := \ln(S_T/K)$, with S_t the underlying price at time t and K the strike price. In our case, as we want the pricing formula for European put options, the payoff, in log-asset price, reads

$$v(y,T) \equiv [-K(e^y - 1)]^+.$$

Notice that for the values $y > 0$, we have that $v(y,T) = 0$ (i.e. the integrand is zero for all $y \in (0,b]$). So, assuming that $a < 0 < b$, integrating from a to b is the same as integrating only up to 0. Defining $w_k = \frac{k\pi}{b-a}$, we observe that V_k can be computed as

$$\begin{aligned} V_k &= \frac{2}{b-a}\int_a^0 K(1-e^y)\sin(w_k(y-a))dy \\ &= \frac{2K}{b-a}\left(\int_a^0 \sin(w_k(y-a))dy - \int_a^0 e^y \sin(w_k(y-a))dy\right). \end{aligned}$$

Following the paper convection, we define two cosine series coefficients ψ_k and χ_k. We will define them as:

$$\psi_k(c,d) = \int_c^d \sin(w_k(y-a))dy,$$

$$\chi_k(c,d) = \int_c^d e^y \sin(w_k(y-a))dy.$$

Therefore, the Fourier-sine coefficients for a vanilla put option are:

$$V_k = \frac{2K}{b-a}\Big(\psi_k(a,0) - \chi_k(a,0)\Big).$$

Now, using basic calculus rules, we can compute ψ_k and χ_k explicitly. For ψ_k, using that $\int \sin(mx+n)dx = -\frac{1}{m}\cos(mx+n) + C$, we can write

$$\begin{aligned}\psi_k(c,d) &= \left[-\frac{1}{w_k}\cos(w_k(y-a))\right]_c^d \\ &= \frac{1}{w_k}\left[\cos(w_k(c-a)) - \cos(w_k(d-a))\right].\end{aligned}$$

For χ_k, using that $\int e^{my}\sin(ny)dy = \frac{e^{my}}{m^2+n^2}[m\sin(ny) - n\cos(ny)]$, we get

$$\chi_k(c,d) = \frac{1}{1+w_k^2}\left[e^y\sin(w_k(y-a)) - w_k e^y\cos(w_k(y-a))\right]_c^d.$$

d. Implementation and Pricing

Implement your formula from Part (c) in Python to price a **European put option**, assuming the underlying stock price follows a log-normal model:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

with parameters:

Parameter	Value
S_0	3
σ	0.3
T	1
K	4
r	0.03

Compare your sine-COS price against the Black–Scholes analytical put price and report the absolute pricing error as a function of the number of terms N.

We implement the Python code and show the table comparing the sine-COS price against the Black–Scholes analytical put price and report the absolute pricing error as a function of the number of terms N.

```
Black-Scholes analytical price: 0.9918191782

    N          SIN price       |SIN - BS|
-----------------------------------------
    4       0.8159724377       1.7585e-01
    8       0.9192207446       7.2598e-02
   16       0.9882986645       3.5205e-03
   32       0.9918190931       8.5116e-08
   64       0.9918191782       3.3307e-16
  128       0.9918191782       3.3307e-16
  256       0.9918191782       3.3307e-16
  512       0.9918191782       3.3307e-16
 1024       0.9918191782       3.3307e-16
```

Figure 2: Errors between analytical put price and sine-COS price for the Black-Scholes as a function of the number of terms N.

We observe a fast convergence of the COS method to the analytical solution, with only $N = 64$, we have reached the analytical solution.

e. CDF Recovery via the Sine Formula

Starting from the sine (SIN) density recovery formula you derived in Parts (a)–(c), extend the derivation by one additional step to obtain a SIN formula for recovering the Cumulative Distribution Function (CDF).

(i) Using your sine series expansion from Part (a), derive the SIN formula for CDF recovery that relies on the characteristic function.

By definition, we know that the CDF of a function $f(x)$ is defined as $F(x) = \int_{-\infty}^{x} f(u)du$. With the bonds $[a, b]$, this becomes

$$F(x) \approx \int_a^x f_2(u)du = \int_a^x \sum_{k=1}^{N} B_k \sin\left(k\pi \frac{u-a}{b-a}\right) du,$$

where $f_2(u)$ is defined in 2(a) and $B_k \approx F_k$, with

$$F_k \equiv \frac{2}{b-a} \Im\left(\phi\left(\frac{k\pi}{b-a}\right) \exp\left(-i\frac{k\pi a}{b-a}\right)\right).$$

Then, we can change the integral and the sum in $F(x)$

$$F(x) \approx \sum_{k=1}^{N} F_k \int_a^x \sin\left(k\pi \frac{u-a}{b-a}\right) du.$$

Now, we compute the integral:

$$\begin{aligned}\int_a^x \sin\left(k\pi \frac{u-a}{b-a}\right) du &= \left[-\frac{b-a}{k\pi} \cos\left(k\pi \frac{u-a}{b-a}\right)\right]_a^x \\ &= \frac{b-a}{k\pi}\left[1 - \cos\left(k\pi \frac{x-a}{b-a}\right)\right],\end{aligned}$$

where we have used that $\cos(0) = 1$. After this computation, we find the SIN formula for CDF:

$$F(x) \approx \frac{2}{\pi} \sum_{k=1}^{N} \frac{1}{k} \Im \left(\phi \left(\frac{k\pi}{b-a} \right) \exp \left(-i \frac{k\pi a}{b-a} \right) \right) \left[1 - \cos \left(k\pi \frac{x-a}{b-a} \right) \right].$$

(ii) Implement your SIN-CDF formula in Python and verify its correctness using the log-normal model from Part (d): take $x = \ln(S_T/S_0)$, where S_t satisfies $dS_t = \mu S_t dt + \sigma S_t dW_t$ with $S_0 = 3$, $\sigma = 0.3$, $T = 1$, $r = 0.03$. Compare your SIN-recovered CDF against the analytical log-normal CDF and produce a table of absolute errors in the same format as Table 1 of the original COS paper, showing convergence as the number of terms N increases.

We implement the Python code and show the table comparing the SIN-recovered CDF against the analytical log-normal CDF, and we show the convergence as the number of terms N increases.

N	Max |CDF error|
4	1.37e-01
8	1.59e-02
16	2.50e-05
32	3.67e-11
64	3.67e-11
128	3.67e-11

Figure 3: Maximum error when recovering the SIN CDF from $\phi(w)$ by Fourier-sine expansion

3 Importance Sampling

Suppose we have a portfolio consisting of $N = 1000$ options, all with the same maturity T. Each option is written on a unique underlying stock $S_i(t)$, for $i = 1, 2, \ldots, N$. At maturity T, each option pays one dollar if $S_i(T) > K_i$; otherwise the payoff is zero. The total portfolio value at maturity is:

$$L_T = \sum_{i=1}^{N} \mathbf{1}_{\{S_i(T)>K_i\}}$$

where $\mathbf{1}_{\{A\}}$ is the indicator function which equals 1 when A is true and 0 otherwise. Each stock price follows, under the risk-neutral measure:

$$\frac{dS_i(t)}{S_i(t)} = r\,dt + \sigma_{i,1}\,dW_t + \sigma_{i,2}\,dB_t^i,$$

where $W(t)$ is a common stochastic factor shared by all stocks, and $B^1(t), B^2(t)$ $, \ldots, B^N(t)$ are independent, stock-specific idiosyncratic factors. $W(t)$ and $B^i(t)$ are independent standard Brownian motions for all i. **Model parameters:**

Parameter	Value
N	1000
$S_i(0)$	1 for all i
r	0.01
$\sigma_{i,1}$	0.8 for all i
$\sigma_{i,2}$	0.6 for all i
T	1
K_i	2 for all i

We are interested in the high quantiles of L_T. The α-quantile is the smallest value q_α such that $\mathbb{P}(L_T \geq q_\alpha) \leq \alpha$.

a. Plain Monte Carlo
Implement an **exact Monte Carlo simulation** of L_T in Python and use it to estimate the **95% quantile** of L_T.
Show in a **table and a figure** that as the number of simulations increases, the estimated 95% quantile converges.

Before implementing the simulation in Python, we compute the SDE exactly to find the dynamics of S_t. In order to do it, we notice that the stock price follows, under the risk-neutral measure:

$$\frac{dS_i(t)}{S_i(t)} = r\,dt + \sigma_{i,1}\,dW_t + \sigma_{i,2}\,dB_t^i,$$

which gives us some intuition to compute the differential of $f(t, S_i) = \ln S_i(t)$. Using the rules of Ito's calculus for a function $f(t, X_t)$, we know

that $df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2$. Then, for our function $f(t, S_t)$ we have that

$$d(\ln S_i(t)) = \frac{1}{S_i}dS_i + \frac{1}{2}\left(-\frac{1}{S_i^2}\right)(dS_i)^2.$$

Now, we only need to compute $(dS_i)^2$, as dS_i is already given. Rewriting, we get that

$$dS_i = rS_i dt + \sigma_{i,1}S_i dW_t + \sigma_{i,2}S_i dB_t^i.$$

Then, we can compute $(dS_i)^2$:

$$(dS_i)^2 = \sigma_{i,1}^2 S_i^2 dt + \sigma_{i,2}^2 S_i^2 dt = (\sigma_{i,1}^2 + \sigma_{i,2}^2)S_i^2 dt,$$

where we have used that W_t and B_t^i are independent Brownian motions, and therefore $dW_t \cdot dB_t^i = 0$. We rewrite the $d(\ln S_i)$ expression:

$$d(\ln S_i(t)) = \left(r - \frac{1}{2}(\sigma_{i,1}^2 + \sigma_{i,2}^2)\right)dt + \sigma_{i,1}dW_t + \sigma_{i,2}dB_t^i.$$

Now, integrating between 0 and T, and then, we exponentiate and get

$$S_i(T) = S_i(0)\exp\left(\left(r - \frac{1}{2}(\sigma_{i,1}^2 + \sigma_{i,2}^2)\right)T + \sigma_{i,1}W_T + \sigma_{i,2}B_T^i\right).$$

After having obtained this expression, we implement the Python code and show the figure and the table for the results.

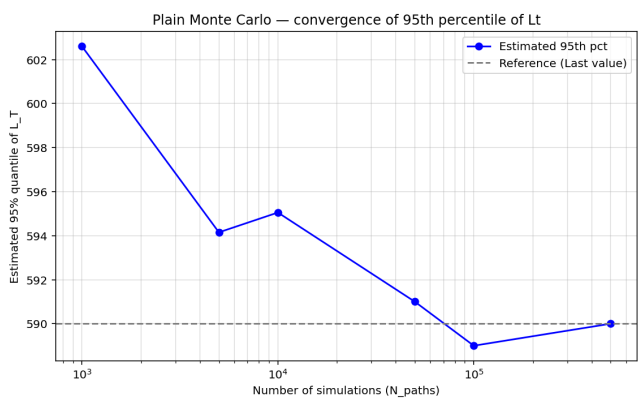

Figure 4: Plain Monte Carlo - convergence of 95th percentile.

N_paths	95th pct
1000	602.60
5000	594.15
10000	595.05
50000	591.00
100000	589.00
500000	590.00

Figure 5: Table with number of simulations and the 95th percentile.

b. Importance Sampling: Mean Shift Using the importance sampling identity: for two strictly positive densities p and q,

$$\int f(x)p(x)\,dx = \int f(x)\frac{p(x)}{q(x)}q(x)\,dx,$$

implement an importance sampling simulation in Python such that, at maturity T, the common factor $W_T \sim \mathcal{N}(1.5, T)$ under the auxiliary measure q.

- Estimate the 95% quantile of L_T under the original measure p, adjusting by the likelihood ratio p/q.
- Show in a **table** and a **figure** that the estimated quantile converges as the number of simulations increases.
- Verify that fewer simulations are needed to achieve the same accuracy as in Part (a).

In (a), we had that under the original measure p, $W \sim N(0, T)$. However, we shift the mean of the common factor so that under the auxiliary measure q, we have $W \sim N(1.5, T)$. This is done to estimate better tail events. Now, because we changed the distribution we must re-weight the outputs using the ratio p/q. Note that the B_t^i terms were not modified, so this ratio will be 1 in their case. Now, we use that $p(w)$ and $q(w)$ are the densities of two Gaussian random variables with means and variances (0,T) and (1.5,T), respectively. Then, we can rewrite the ratio p/q as

$$\begin{aligned}
\frac{p(w)}{q(w)} &= \frac{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{w^2}{2T}\right)}{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(w-1.5)^2}{2T}\right)} = \exp\left(\frac{-w^2 + (w-1.5)^2}{2T}\right) \\
&= \exp\left(\frac{-w^2 + w^2 - 3w + 2.25}{2T}\right) \\
&= \exp\left(\frac{-3w + 2.25}{2T}\right).
\end{aligned}$$

As the maturity time is $T = 1$, we get that $\frac{p(w)}{q(w)} = \exp(-1.5w + 1.125)$. After these computations, we proceed to implement the code and show the results.

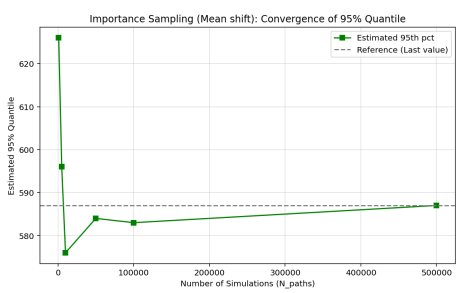

Figure 6: Importance Sampling (Mean Shift) - convergence of 95th percentile.

N_paths	95th pct
1000	626.00
5000	596.00
10000	576.00
50000	584.00
100000	583.00
500000	587.00

Figure 7: Table with number of simulations and the 95th percentile.

To verify the efficiency of the mean-shifted importance sampling method, we compare its convergence to the plain Monte Carlo method. In plain Monte Carlo 3(a), the 95% quantile implies that approximately only 5% of the simulated paths cross the threshold of interest. The desired samples are rare and, consequently, the estimator suffers from high variance, requiring a large number of simulations) to guarantee a stable quantile estimate.

However, by shifting the mean of the common factor W_T from 0 to 1.5 under the new measure q, we force the center of the distribution to be in our target region (in the tail). As a consequence, we have more paths or simulations in our region of interest, around the 95% quantile. Therefore, we need fewer simulations to achieve the same accuracy as plain Monte Carlo method.

c. Importance Sampling: Mean and Variance Shift Now implement an importance sampling simulation whereby, at maturity T, the common factor satisfies:

$$W_T \sim \mathcal{N}(1.5, 2T)$$

under the auxiliary measure q (i.e. the variance is doubled relative to the original measure).

- Estimate the 95% quantile of L_T under the original measure p, adjusting by the likelihood ratio p/q.
- Show in a table and a figure that the estimated quantile converges as the number of simulations increases.
- Investigate whether this variance-shifted scheme requires fewer simulations to achieve the same accuracy as in Part (a), and comment on your findings.

Similar to 3(b), we need to compute the ratio p/q, as we change from the original measure p, to a new auxiliary measure q, where the common factor satisfies $W \sim N(1.5, 2T)$. Now, we use that $p(w)$ and $q(w)$ are the densities of two Gaussian random variables with means and variances (0,T) and (1.5,2T), respectively. Then, we can rewrite the ratio p/q as

$$\begin{aligned}
\frac{p(w)}{q(w)} &= \frac{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{w^2}{2T}\right)}{\frac{1}{\sqrt{4\pi}} \exp\left(-\frac{(w-1.5)^2}{4T}\right)} = \sqrt{2} \exp\left(\frac{-2w^2 + (w-1.5)^2}{4T}\right) \\
&= \sqrt{2} \exp\left(\frac{-2w^2 + w^2 - 3w + 2.25}{4T}\right) \\
&= \sqrt{2} \exp\left(\frac{-w^2 - 3w + 2.25}{4T}\right).
\end{aligned}$$

As the maturity time is $T = 1$, we get that $\frac{p(w)}{q(w)} = \sqrt{2}\exp\left(\frac{-w^2-3w+2.25}{4}\right)$. After these computations, we proceed to implement the code and show the results.

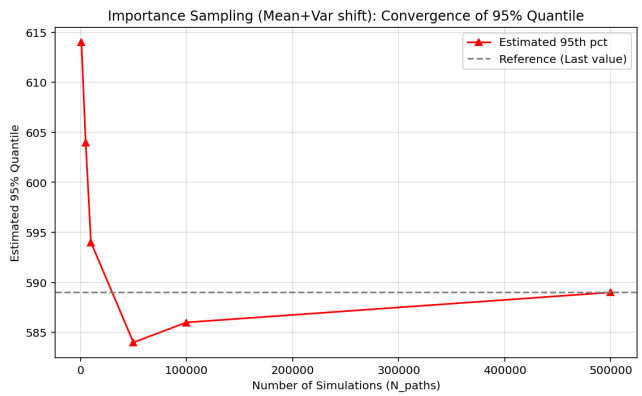

Figure 8: Importance Sampling (Mean Shift) - convergence of 95th percentile.

N_paths	95th pct
1000	614.00
5000	604.00
10000	594.00
50000	584.00
100000	586.00
500000	589.00

Figure 9: Table with number of simulations and the 95th percentile.

Increasing the variance of the common factor means that the sampling distribution becomes more spread out. This leads to more simulated paths in the extreme right tail. This guarantees that the ratio p/q is well-defined because $p/q \to 0$ as $w \to \infty$, but is not helpful at all for computing the 95% percentile. In Part 3(b), the mean shift moves the sampling distribution closer to the region around the 95% threshold. This concentrates the simulations in the area that is more important for estimating the quantile. However, in 3(c), when we double also the variance, the distribution becomes flatter. As a result, too many simulations fall in the extreme tail, far beyond the 95% threshold.

Because of this, a less number of effective weighted samples remain near the 95% quantile. This makes the empirical CDF around that point less stable and less efficient. Therefore, the variance-shifted importance sampling method might require more simulations (and will be more expensive) to obtain a stable estimate of the 95% quantile compared to the mean-shifted method.